

Course: B. Tech.

Semester: III

Branches: Mechanical Engineering / Mechanical Engineering (Sandwich) /

Automobile Engineering/ Production Engineering

Subject Code & Name: BTMES304 Materials Science and Metallurgy

Max Marks: 60

Date: 06/07/2024

Duration: 3 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) What are the three main types of crystal structures? Explain any two of them in brief using schematic diagrams.	L1/CO1	6
B) Give the classification of various crystal defects and explain any two of those in detail using neat sketches.	L1/CO1	6
C) Compare Brinell Hardness Test with Rockwell Hardness Test.	L2/CO2	6
Q.2 Solve Any Two of the following.		12
A) Explain the Gibbs phase rule. What are the Hume-Rothery's rules of solid solubility?	L2/CO3	6
B) Draw a neat diagram for Fe-Fe ₃ C equilibrium diagram. Show all the phases, transformations and important temperatures on it.	L2/CO3	6
C) Classify steels on the basis of amount of carbon and grain coarsening characteristics. Explain each type in brief.	L1/CO3	6
Q. 3 Solve Any Two of the following.		12
A) Explain the mechanism of removal of heat during cooling with the help of schematic diagrams/graphs. Give the requirements of an ideal quenchant.	L2/CO4	6
B) Differentiate clearly between annealing and normalizing heat treatments. Show the temperature range of these treatments on the equilibrium diagram.	L2/CO4	6
C) Describe nitriding type of surface treatment w.r.t. process details, mechanism and schematic diagram of gas nitriding, and advantages.	L3/CO4	6

Q.4 Solve <i>Any Two</i> of the following.		12
A) Compare macroscopy with microscopy w.r.t. definition, purposes, specimen preparation, merits, and demerits.	L1/CO5	6
B) Describe the procedure for specimen preparation for microscopy.	L1/CO5	6
C) Discuss the spark test w.r.t. purpose, mechanism of generation of spark, and the spark patterns for low carbon and high carbon steel.	L2/CO5	6
Q. 5 Solve <i>Any Two</i> of the following.		12
A) Explain refinement of grain size as the strengthening mechanism w.r.t. Hall-Petch equation, advantages, and commercial examples.	L1/CO6	6
B) Discuss age/precipitation hardening process w.r.t. basic principle, conditions, and steps.	L2/CO6	6
C) Describe the procedure of dye penetrant inspection using schematic diagrams.	L1/CO6	6

***** End *****